

Generation Interconnection System Impact Study Report

For Queue Project Q20-1375

Red Mesa 100 kV

March 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1375. The Interconnection Customer proposes a 150.0 MW solar photovoltaic facility interconnecting at the Red Mesa 100 kV substation in the Ohio area. The request was received on January 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 483 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$20,436,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Sta F to Red Mesa 100kV
Contingency: Barrett to Wolfe 500kV line
Loading: 164.9% (Rating: 107 MVA)
DFAX: 61.9%
Pre-project loading: 77.9%
2. Facility: Elma to Red Mesa 100kV
Contingency: Red Mesa to Sta F 100kV
Loading: 127.3% (Rating: 113 MVA)
DFAX: 100.0%
Pre-project loading: 4.9%
3. Facility: Elma to Kycee 100kV
Contingency: Sta F to Red Mesa 100kV
Loading: 111.0% (Rating: 120 MVA)
DFAX: 100.0%
Pre-project loading: 14.0%
4. Facility: Little Chute to Talenenergy Martins Crk Llc Harrisburg 345kV
Contingency: Ft Humug to Coronet 500kV
Loading: 102.1% (Rating: 2,249 MVA)
DFAX: 5.9%
Pre-project loading: 101.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Sta F to Red Mesa 100kV	Rebuild line with higher-capacity conductor	\$6,756,000	100% (\$6,756,000)
Elma to Red Mesa 100kV	Rebuild line with higher-capacity conductor	\$4,519,000	100% (\$4,519,000)
Elma to Kycee 100kV	Rebuild line with higher-capacity conductor	\$9,043,000	100% (\$9,043,000)
Little Chute to Talenenergy Martins Crk Llc Harrisburg 345kV	Reconductor line	\$10,468,000	1% (\$118,000)
	Total allocated to project		\$20,436,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.